

3D Scatter Plot Projection Project

Project Overview:

This project displays randomly generated 3D points on an HTML5 Canvas. The user can rotate the point cloud using **Yaw**, **Pitch**, and **Roll** controls. The project also allows switching between **Orthographic Projection** and **Perspective Projection** to view the points in different ways.

Workflow:

1. Generate random 3D points.
2. Store the points in a JSON file.
3. Load the points into JavaScript.
4. Convert the rotation angle from degrees to radians.
5. Apply **Pitch (X-axis)**, **Yaw (Y-axis)**, and **Roll (Z-axis)** rotations.
6. Project the 3D points into 2D.
7. Draw the points on the canvas.
8. Repeat the process for smooth animation using Auto Rotate.

Random Number Generation

Each point contains three values:

- **X coordinate**
- **Y coordinate**
- **Z coordinate**

The values are generated randomly within a fixed range (approximately **-400 to +400**) to create a 3D scatter of points.

Rotation Formulas

Pitch (X-axis)

$$y' = y \cos\theta - z \sin\theta$$

$$z' = y \sin\theta + z \cos\theta$$

Yaw (Y-axis)

$$x' = x \cos\theta + z \sin\theta$$

$$z' = -x \sin\theta + z \cos\theta$$

Roll (Z-axis)

$$x' = x \cos\theta - y \sin\theta$$

$$y' = x \sin\theta + y \cos\theta$$

Degree to Radian Conversion

Formula

$$\text{Radians} = \text{Degrees} \times \pi / 180$$

Orthographic Projection

Orthographic Projection ignores the depth (Z-axis).

Formula

$$x' = x$$

$$y' = y$$

Perspective Projection

Perspective Projection considers depth and gives a realistic 3D view.

Formula

$$\text{Scale} = f / (f + z)$$

$$x' = x \times \text{Scale}$$

$$y' = y \times \text{Scale}$$

Overall Implementation

The project first generates random 3D points and stores them in a JSON file. These points are loaded into JavaScript. The selected rotation angles are converted into radians and applied using the Pitch, Yaw, and Roll formulas. After rotation, the points are projected into 2D using either Orthographic or Perspective Projection and displayed on the HTML5 Canvas. When Auto Rotate is enabled, the points rotate continuously and the scene is updated using `requestAnimationFrame()` for smooth animation.